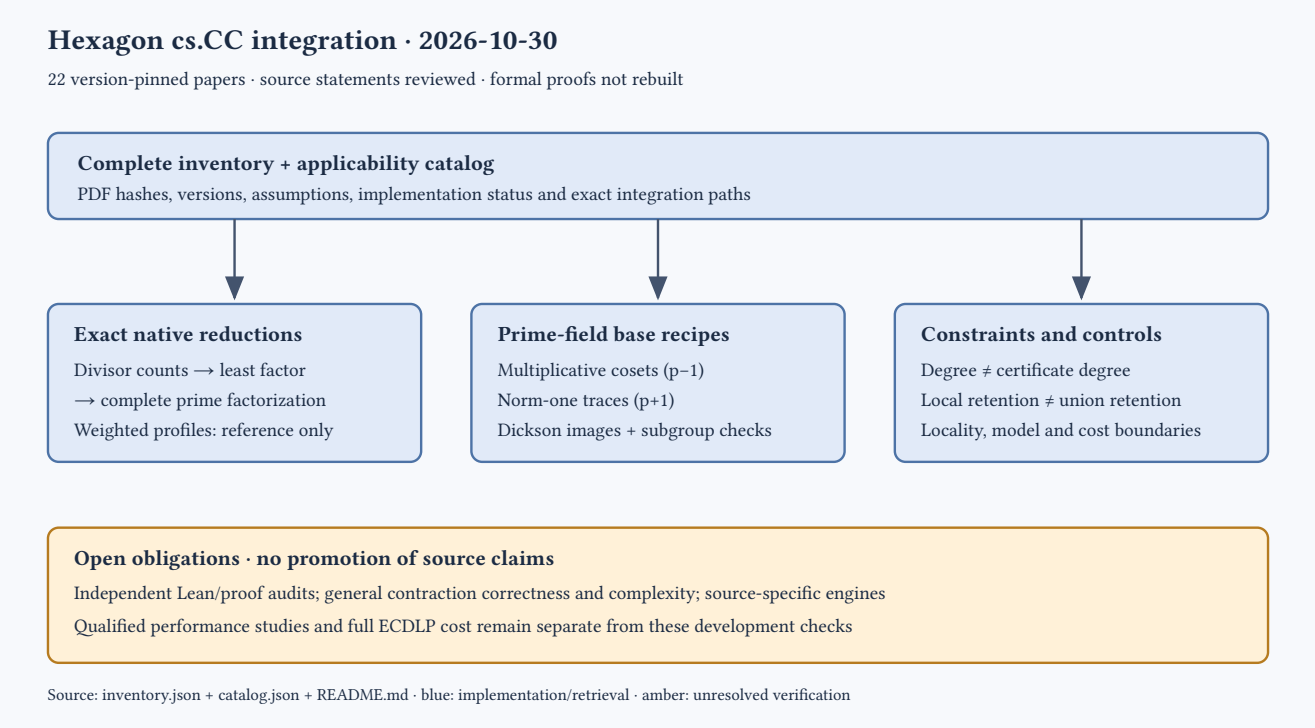
1. Hexagon computational-complexity integration

2026-10-30 · isogeny-labs/crypto, with sibling integrations

This report records a source review and native software additions. It reports no benchmark, new theorem, independently checked source proof, ECDLP result or scientific state transition. The complete subject snapshot contains 22 works, including cross-listings and the three current v2 records. All primary PDFs were retrieved and hashed. Statements and relevant limitations were reviewed; complete proofs and Lean developments were not independently rechecked.



1.1. Executable work and scope

The native tensor library now provides exact divisor-prefix counting, smallest-factor search and complete factorization, plus an exhaustive local-cost profile reference. The counter is the existing OpenAI-based implementation, whereas the Hexagon factoring note re-proves an Anthropic engine. This is an implementation of the reduction, not its source-specific asymptotic proof.

The factor-base factory adds arbitrary bounded-order multiplicative cosets, norm-one torus traces and Dickson images. Exact supplied-generator checks, canonical coordinates, deduplication, subgroup filtering and archive replay remain explicit. These recipes do not by themselves supply a faster PDP solver.

The native catalog checks complete snapshot coverage and retrieves source claims together with their assumptions and implementation status. Finite controls reproduce the Boolean three-generator and affine-union constructions; they do not independently prove universal degree or asymptotic lower bounds.

1.2. Remaining obligations

Independent formal/source audits, the general correctness and cost proof for contraction, the Anthropic engine port, subexponential weighted contraction and fast Subset Sum/3SUM implementations remain unperformed. Correctness tests do not establish runtime improvements. Existing performance/frontier graphs receive no new data. Full commands, observations and failed attempts are in VALIDATION.md; the complete discussion is in README.md.

1.3. Every source and its integration

1.3.1. What Does the Refutation of the Exponential Time Hypothesis Mean for Learning Theory?

hexagon:2610.00613v1 · Xiaoyu Li

Locality, description length, accuracy and coefficient representation govern the transfer from counting to learning and exact inference. Fixed-width SETH and unbounded-width CNF-SETH are different hypotheses.

Scope: Bound the bits read by each local factor; Include precision and sample/formula length; Do not infer a bound for arbitrary dense models.

Integration: reference_backend. crates/hashed-ecdl-tensor/src/weighted.rs; src/ml_dataset

Evidence: Imported ETH counting theorems and written reductions; no local formal verification. Local formal verification: not_run.

PDF SHA-256: 0302872eb0536c1954fa0379fe94be26408cb7f76e5556b512bdfcc52dc91303

1.3.2. The Euclidean algorithm is not optimal among uniform processes

hexagon:2610.00604v1 · Jason Davies

Reusable Euclidean prefixes reduce remainder-only derivation length. Uniform-process existence does not by itself supply a faster finite-program gcd implementation.

Scope: The cost counts remainder calls, not selection or bit complexity; The Main Conjecture for algorithms remains open.

Integration: research_constraint. research/hexagon_cscc_20261030/README.md; src/utils

Evidence: Main remainder-call bounds reported formalized; best-uniform-process theorem imported. Local formal verification: not_run.

PDF SHA-256: c86443a64680722b8dc78d6c37b62720854d2d468fc5b640bd840844e31c161f

1.3.3. Deterministic Factoring in Time $N^{o(1)}$

hexagon:2610.00600v1 · Isaac Rudich, Louis-Martin Rousseau

Count nontrivial divisors by binary prefix, choose the least feasible prefix, and repeat smallest-factor extraction for complete factorization.

Scope: One fixed machine in the asymptotic statement; The local counter is OpenAI-based, not the paper's Anthropic engine; The stronger sqrt-log-log rate is a conjecture.

Integration: implemented_reduction. crates/hashed-ecdl-tensor/src/factoring.rs; crates/hashed-ecdl-tensor/src/bin/tensor-factor.rs

Evidence: Theorem 1 reported Lean-checked; Theorem 2 is a written corollary; source proof not rebuilt locally. Local formal verification: not_run.

PDF SHA-256: e57f24481d2474a6f2c342d86312572f024710835ca583beb31e2bb1f9ceae40

1.3.4. Subexponential Weighted Counting and Exact Optimisation

hexagon:2610.00594v1 · Deep Bhattacharjee

Use generating-polynomial coefficients to recover cost distributions and optima from local weighted counting, rather than replacing ring operations by min-plus operations.

Scope: Commutative Q-algebra for the source interpolation; Fixed locality and linear factor density in Theorem 5.3; Charge coefficient precision and degree; handle truncation explicitly.

Integration: reference_backend. crates/hashed-ecdl-tensor/src/weighted.rs

Evidence: Written proof and source reproduction; local implementation is exhaustive and asserts no source runtime. Local formal verification: not_run.

PDF SHA-256: c3cf905b7b85085d63d8aa41082eadfcc0bf2fbea390ee07e1457f51d4cf9009

1.3.5. On counting satisfying assignments via polynomial separation

hexagon:2610.00574v1 · Scott Kominers

Separate the counting recovery interface from the construction supplying polynomial separation. Reconstruct exactly from Taylor data and Hermite interpolation.

Scope: Account for construction, query and interpolation costs; Qualitative epsilon-dependent constants are not concrete benchmark parameters.

Integration: existing_backend_context. crates/hashed-ecdl-tensor/src/contraction/deformation.rs; crates/hashed-ecdl-tensor/src/contraction/recursive.rs

Evidence: Expository qualitative proof; independent implementation audit remains open. Local formal verification: not_run.

PDF SHA-256: 4d23c13745bb213808942364eda9e8972ab93bc143eeae0818c9c0a0a1e14552

1.3.6. Some exposition of “The Exponential Time Hypothesis Is False”

hexagon:2610.00572v1 · Ryan O'Donnell

A low-rank correction to commuting SWAP operators explains the separation construction. The exposed cubic-graph independent-set counter is Las Vegas, with exact answers and randomized labeling.

Scope: Quantum notation describes classical linear algebra; Randomized labeling and claimed derandomization are separate; Do not label an elimination-order limitation a general algorithm lower bound.

Integration: existing_backend_context. crates/hashed-ecdl-tensor/src/contraction/deformation.rs; crates/hashed-ecdl-tensor/src/weighted.rs

Evidence: Exposition; no local proof replay or new recursion implementation. Local formal verification: not_run.

PDF SHA-256: 85b54ea5dce595d2ae7df8d660186bfb6ce3a065a72e47e2e84d607c9e44dbb2

1.3.7. A $2^{o(n)}$ Time Algorithm for 3-SAT

hexagon:2610.00577v1 · Anthropic

Recursive algebraic network decompositions shrink the exponent for exact k-CNF counting. The variable, clause and width parameters and the universal-machine quantifiers must remain explicit.

Scope: Source constants force astronomical crossover; Fixed-k ETH and fixed-k SETH consequences do not refute every fine-grained hypothesis; The Hexagon factoring note re-proves only part of this engine.

Integration: research_constraint. crates/hashed-ecdl-tensor/src/factoring.rs; research/hexagon_cscc_20261030/README.md

Evidence: Anthropic reports Lean and independent-kernel checks; not rerun in this session. Local formal verification: not_run.

PDF SHA-256: 85034e06e1d058aa4ed7d895c226ae755ebd3a5c6614b07ba14981915c4f7360

1.3.8. The rank of 3×3 matrix multiplication over F2 is 23

hexagon:2610.00548v1 · Tejasvi Singh Tomar

The reported exact bilinear rank of 3x3 matrix multiplication over F2 is 23; exhaustive completion produces independently checked certificates.

Scope: Bilinear rank over F2 is not wall-time optimality; Do not transfer an F2 rank bound to arbitrary fields or nonlinear algorithms; Compiled-code and pure-kernel checks have different trust surfaces.

Integration: research_constraint. src/cryptanalysis/matrix_f5_f2.rs; src/cryptanalysis/f4_rref_tests.rs

Evidence: Two Lean verification routes reported; neither rebuilt locally. Local formal verification: not_run.

PDF SHA-256: 57165c9d24af5baa4747eea6eabc98abd0ad523564faebf7981e86251b5bb90e

1.3.9. A degree-three local-weight cap and an improved relevant-variable bound

hexagon:2610.00514v1 · Ricardo Pinho

A degree-three local-degree potential cap improves a relevant-variable bound using exact rational certificates and an explicitly bounded infinite tail.

Scope: Uses an imported ten-variable classification; Numerical certificate checking is not the full universal proof; Fourier/Boolean degree is not a Groebner solving degree.

Integration: research_constraint. src/cryptanalysis/boolean.rs; research/hexagon_cscc_20261030/README.md

Evidence: Two exact-arithmetic checkers reported; no proof-assistant verification claimed. Local formal verification: not_run.

PDF SHA-256: e64a38406fd32a0d6f9fc05461b760d0d78e3b728cf1bb852a2ad3dd467ada50

1.3.10. Polynomial Fault Detection Without Bounded Width: Exact Reachable Sets in Priority Controllers

hexagon:2610.00503v1 · Idris Ali Shaik

Exact reachable images can make a specified fault-detection problem tractable despite large native width. Changing the fault type, optimization objective or counting task changes complexity.

Scope: Only the specified bridge fault and guarded-decoder class; Do not transfer detection algorithms to minimum witnesses or counts; Retain full native CNF encoding and witness replay.

Integration: research_constraint. src/cryptanalysis/sat.rs; src/cryptanalysis/prepared_sat_control.rs

Evidence: Image statements reported formalized; broader algorithms and hardness are written arguments. Local formal verification: not_run.

PDF SHA-256: fc78672c5fd93457caa65d778b6b8eca4b6dde4308c907e58f27bf50ddfaf4cb

1.3.11. Unconstrained detection of productive autocatalytic cores is NP-complete

hexagon:2610.00390v1 · Michael Mislan

Literal reactant/product participation and signed-flow feasibility must be checked separately; cancellation in a net matrix does not preserve the original admissibility predicate.

Scope: Binary literal-source encoding; Unconstrained structural possibility, not kinetics; Hardness needs the explicit model-preserving reduction.

Integration: cross_domain_constraint. research/hexagon_cscc_20261030/README.md

Evidence: NP-completeness reported Lean-verified; no biochemical solver integration. Local formal verification: not_run.

PDF SHA-256: d0a4463f5ff4395fc7bcc69f072c5a5578be2f3e142ff3840cc2fae5467a7096

1.3.12. Deciding D-stability is coNP-hard

hexagon:2610.00368v1 · Michael Mislan

Universal positive-diagonal stability is different from testing a single scaling. The hardness reduction tracks integer magnitudes and polynomial binary output length.

Scope: Weak hardness from binary PARTITION; coNP-hard is not coNP-complete; Numerical eigenvalue samples do not certify all diagonal scalings.

Integration: cross_domain_constraint. research/hexagon_cscc_20261030/README.md

Evidence: Arithmetic and promise dichotomy reported formalized; total complexity argument partly conventional. Local formal verification: not_run.

PDF SHA-256: 1b2cf8307e1476977d0f8fda9e25ef90b806a0a5064a588e961b1fd4718da75f

1.3.13. A Three-Polynomial Counterexample to the Generalized Effective Hypercube Nullstellensatz

hexagon:2610.00327v1 · YURAN XIU

Three low-degree Boolean generators can require unbounded reduced-product certificate degree even when their product is zero. Selector gadgets preserve hard faces.

Scope: Real coefficients and Boolean-cube multilinearization; Does not contradict the two-generator theorem; Input degree alone does not bound certificate or solving degree.

Integration: bounded_control. tools/complexity-catalog/src/controls.rs; src/cryptanalysis/groebner_f4.rs; src/cryptanalysis/f4_fp_tower.rs

Evidence: Unbounded theorem reported Lean-checked; local controls only check finite witness properties. Local formal verification: not_run.

PDF SHA-256: 46867c11115644eb3b98df259fd21398a9ee17b360e8316670cdcd710c9e06ee

1.3.14. A Counterexample to Affine-Subspace Union Stability under Small Deletions

hexagon:2610.00328v1 · YURAN XIU

Deleting a 1/N fraction of each coordinate subspace can delete a central-binomial fraction of their union. Per-component retention is not global coverage retention.

Scope: Highly overlapping affine coordinate subspaces; Local and global denominators must be distinguished; Finite witnesses do not independently prove the all-exponent asymptotic claim.

Integration: bounded_control. tools/complexity-catalog/src/controls.rs; tools/factor-base/src/recipe.rs; src/cryptanalysis/ic_framework/subspace_select.rs

Evidence: Arbitrarily-large counterexample reported Lean-checked; finite overlap controls implemented locally. Local formal verification: not_run.

PDF SHA-256: 71f263e05a17fa29ab3017e7f9d2bcc77e6a6a529d71e0b06dfddda0231759c1

1.3.15. An Approximate Solution to the Minimum Vertex Cover Problem: The Salvador Algorithm

hexagon:2610.00337v1 · Frank Vega

An ensemble's per-instance certificate must be separated from its conjectured universal approximation guarantee. Keep every heuristic arm and exact lower-bound witness.

Scope: The universal 3/2 ratio remains conjectural; The 2*tau-r bound has different scope; Test-suite success is not a universal approximation proof.

Integration: research_constraint. src/cryptanalysis/ic_framework/planner.rs; research/hexagon_cscc_20261030/README.md

Evidence: Written bounds and empirical suite; stronger guarantee explicitly conjectured. Local formal verification: not_run.

PDF SHA-256: 1db96e3ef4968d3372ccd11e32148738b5d19382c778055fc64f86266157bfe3

1.3.16. The Exact Junta Bound for Degree-Two Boolean Functions on q-ary Hamming Cubes

hexagon:2610.00301v2 · YURAN XIU

Boolean functions of Hamming degree at most two over a q-symbol alphabet have at most max(4,q+1) relevant coordinates; the larger-than-four case has selector structure.

Scope: Hamming/junta degree, not ordinary univariate polynomial degree; Boolean output range; The selector conclusion does not make every four-junta two-query.

Integration: research_constraint. src/cryptanalysis/boolean.rs; crates/hashed-ecdl-tensor/src/weighted.rs

Evidence: Exact bound and selector structure reported formalized; no local proof replay. Local formal verification: not_run.

PDF SHA-256: b418c47224062e98a80c590f4a7fcd54703b8725d9436fae306e4d44dd39a5d6

1.3.17. Set Cover by Cyclic Rotations Is NP-Complete

hexagon:2610.00307v2 · YURAN XIU

Restricting set cover to rotations of one pattern does not guarantee tractability. The reduction's full indicator word and binary budget are part of the complexity claim.

Scope: Explicitly listed pattern; All rotations and empty patterns included; Cyclic symmetry alone does not imply a fast exact coverage selector.

Integration: research_constraint. src/cryptanalysis/ic_framework/subspace_select.rs; research/hexagon_cscc_20261030/README.md

Evidence: Reduction, machine-level encoding and NP membership reported formalized. Local formal verification: not_run.

PDF SHA-256: 29362165430615aee9a3981d297220b75ac5a282e5b127c22c113c90830ae20e

1.3.18. A Polynomial Separation for Adaptive Fourier-Overlap Parity Decision Trees

hexagon:2610.00308v2 · YURAN XIU

Maximizing Fourier-support overlap can force polynomial-depth parity trees under every adaptive tie policy, even when unrestricted parity trees have logarithmic depth.

Scope: Lower bound concerns this support-overlap query rule only; Characters must be nonconstant on the current affine domain; A singleton Fourier support need not mean a constant function.

Integration: research_constraint. src/cryptanalysis/boolean.rs; src/cryptanalysis/ic_framework/planner.rs

Evidence: Adaptive separation reported Lean-checked; no general parity-tree lower bound claimed. Local formal verification: not_run.

PDF SHA-256: a744def9cfe67d4c1b0231b203660eceb5f2dc03abfd7db0a788e3ebdf083665

1.3.19. Subset Sum in time O(2^{0.4877n})

hexagon:2610.00261v1 · Annanay Kapila

Phase-summed aliases and a pattern-selected checksum-filter family refine a randomized Subset Sum algorithm; checksums must not silently discard repeated representations.

Scope: Polynomially bounded input bit lengths; Uniform randomized word-RAM algorithm with bounded error; Arb certificates verify numerical inequalities, not the entire algorithm.

Integration: research_constraint. src/cryptanalysis/ic_framework/large_prime.rs; research/hexagon_cscc_20261030/README.md

Evidence: Written proof importing an earlier algorithm plus numerical certificates; fast Subset Sum engine not ported. Local formal verification: not_run.

PDF SHA-256: 1fa891e6ce4a32a171977e9f00c6d24f6bf54f6586d5e16a8d5e09bdf9caca32

1.3.20. Deterministic Integer 3SUM in O(n^1.999017) Time via a Sharper Binomial-Tail Analysis

hexagon:2610.00221v1 · 宇轩 徐

Keeping the binomial tail in a preprocessing analysis gives a smaller 3SUM exponent within an inherited thin-matrix/Exact-Triangle framework.

Scope: Polynomially bounded integers and logarithmic word size; Three distinct input indices, even when values coincide; Include preprocessing, prime selection, query and witness costs.

Integration: research_constraint. src/cryptanalysis/ic_framework/planner.rs; research/hexagon_cscc_20261030/README.md

Evidence: Exact rational interval certificates reported; improved runtime not Lean-formalized or locally implemented. Local formal verification: not_run.

PDF SHA-256: c3f7f56b1ace91a623f6651aaef45cf4ada0af7bd7687c241a5e42ca5efdaff5

1.3.21. A Verified Constructive Reduction of the Cook-Levin Theorem in Lean 4: Bridging the Gap Between Complexity Theory and Formal SAT Encodings

hexagon:2610.00152v1 · Jonathan *f*(n) Reed

A concrete machine-to-CNF reduction needs injective variable indexing and encoding soundness. A high-level theorem alone does not certify an emitted solver instance.

Scope: Check indexing domains, boundary times, initialization, transitions and acceptance; Record auxiliary-variable growth; The claimed complete formalization needs a statement and axiom audit.

Integration: research_constraint. src/cryptanalysis/sat.rs; src/cryptanalysis/semaev_sat.rs; crates/hashed-ecdl-tensor/src/network.rs

Evidence: Lean snippets and claimed formalization in paper; complete source build not independently reproduced. Local formal verification: not_run.

PDF SHA-256: 6f4a167d35e3f32455160a0ef1bddb55d3400bd9ba1807e338948367ea201cbb

1.3.22. Generalized Bawo: A Finite-State Model and an EXPTIME Upper Bound

hexagon:2609.00122v1 · Isaac kaimfa

A lap cap guarantees finite moves; an explicit state bound and WIN/LOSS/DRAW attractor give an EXPTIME upper bound for the defined game model.

Scope: Binary cap encoding and successor-generation cost; Termination guards are part of semantics; An upper bound is not hardness and does not classify different real-world rules.

Integration: cross_domain_constraint. research/hexagon_cscc_20261030/README.md

Evidence: Written finite-state upper bound; no formal or hardness result claimed. Local formal verification: not_run.

PDF SHA-256: 81e9a6c2c73535e019ee194a5ca9c96f7577a27fd499a1e394f33d05088451bb